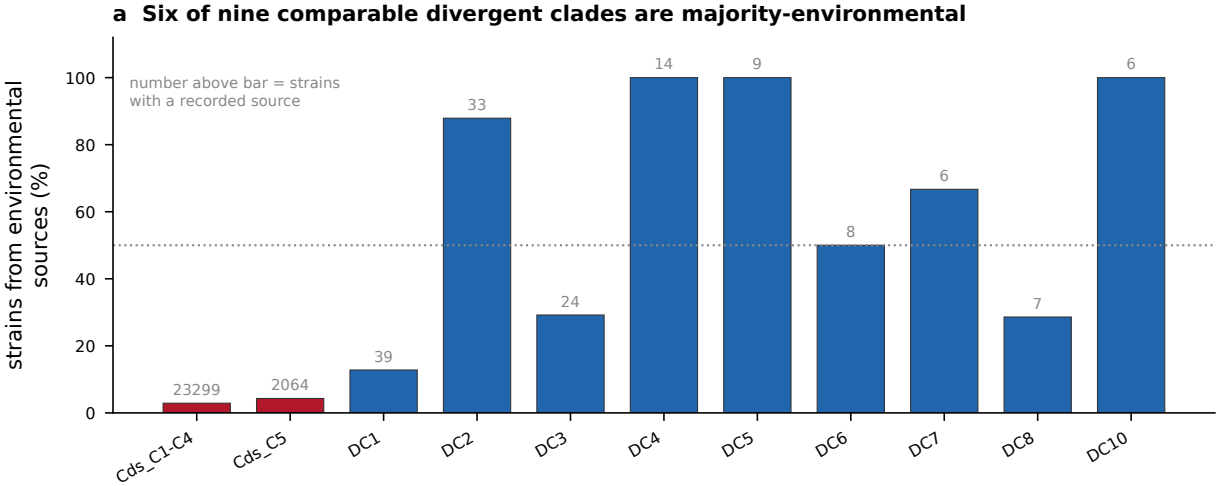
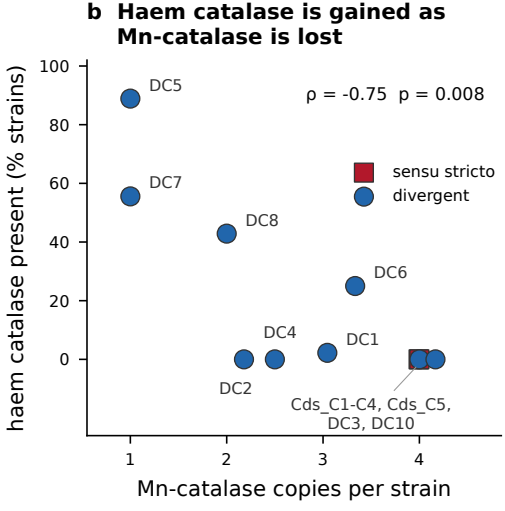
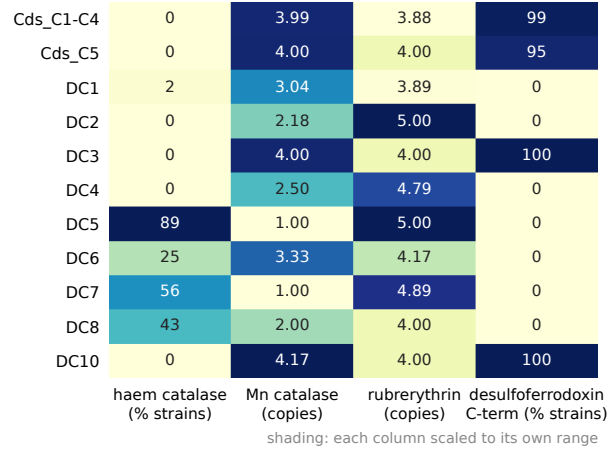

a Six of nine comparable divergent clades are majority-environmental
strains from environmental sources (%)
number above bar = strains with a recorded source
100
80
60
40
20
0
23299
2064
39
33
24
14
9
8
6
7
6
Cds_C1-C4
Cds_C5
DC1
DC2
DC3
DC4
DC5
DC6
DC7
DC8
DC10


b Haem catalase is gained as Mn-catalase is lost
haem catalase present (% strains)
ρ = -0.75 p = 0.008
sensu stricto
divergent
DC5
DC7
DC8
DC6
DC4
DC1
DC2
Cds_C1-C4, Cds_C5, DC3, DC10
100
80
60
40
20
0
1
2
3
4
Mn-catalase copies per strain


c The system is reconfigured, not gained
Cds_C1-C4 0 3.99 3.88 99
Cds_C5 0 4.00 4.00 95
DC1 2 3.04 3.89 0
DC2 0 2.18 5.00 0
DC3 0 4.00 4.00 100
DC4 0 2.50 4.79 0
DC5 89 1.00 5.00 0
DC6 25 3.33 4.17 0
DC7 56 1.00 4.89 0
DC8 43 2.00 4.00 0
DC10 0 4.17 4.00 100
haem catalase (% strains)
Mn catalase (copies)
rubrerythrin (copies)
desulfoferrodoxin C-term (% strains)
shading: each column scaled to its own range